Flood Relief Management System
Project Report

Course: CSC 1051 – Full-Stack Fundamentals

B.Sc. (General) Degree – First Year, Second Semester

University of Sri Jayewardenepura

Group Number: 18

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Contents

1. Introduction 2
2. System Overview 2
Target Users: 2
Technologies Used: 3
Database Structure (Main Tables): 3
Main Backend Functions: 3
Frontend Functionalities: 3
3. Results 4
Main System Pages: 4
Screenshots 5
4. Group Members' Contributions 17

1. Introduction

This project presents the development of a web-based Flood Relief Management System. The main purpose of the system is to support people affected by floods by providing a simple digital platform where they can submit their relief requirements. The system also allows administrators to monitor requests and manage user information.

In Sri Lanka, floods are a common natural disaster that affects many communities. During such situations, collecting accurate information about affected households is important for distributing relief effectively. This system helps organize and store those requests in a structured database.

Objectives of the System:

- Objectives belong to user:
    - Allow affected individuals to register and log in to the system.
    - Provide a form to submit flood relief requests.
    - Enable users to update, view and delete their requests when necessary.
- Objectives belong to administrator:
    - Allow administrators to view users and monitor their relief data in the system.
    - Allow administrators to view detailed information about each separate user.
    - Allow administrator to filter user requests.
    - Allow administrators to remove users from the system.
    - Generate simple summary reports based on system data.

2. System Overview

The Flood Relief Management System is designed as a web application that connects users and administrators through a centralized platform. The system includes authentication, relief request management, and reporting features.

Target Users:

• Affected Person (User) – Can submit and manage relief requests.

• Administrator – Can monitor system data and manage users.

Technologies Used:

- Frontend: HTML, CSS, JavaScript
- Backend: PHP (without frameworks)
- Database: MySQL managed through phpMyAdmin

These technologies were integrated to build a complete full-stack application. HTML and CSS were used to design the user interface, JavaScript was used for client-side interactions, PHP handled server-side processing, and MySQL was used to store system data.

Database Structure (Main Tables):

• users – stores user account information.

• relief_requests – stores flood relief request details.

Main Backend Functions:

- Make a connection between front end and database via connection PHP file.
- User registration, login validation and logout from user dashboard.
- Creating, updating, and deleting relief requests.
- Admin functions for viewing users, generating summaries and logout from admin dashboard

Frontend Functionalities:

- **Form Validation for Relief Requests:**
    - Intercepts the submission of the form with the ID relief form.
    - Checks if all required fields are filled out using the browser's built-in validation.
    - Prevents the form from being sent and alerts the user if any required information is missing.
- **User Registration Handling:**
    - Sets up an event listener for the signupForm submission.

- **Dynamic Deletion Confirmation:**
    - Provides a reusable confirmDeletion function that triggers a browser confirmation pop-up.
    - **Context-Aware Messaging:** It adapts the warning message based on whether you are deleting a "user" or a "relief request."
- **User Detail Navigation:**
    - Provides a viewUserDetails function that redirects the browser to a specific profile page.

3. Results

The completed system provides several functional interfaces for both users and administrators. After logging in, users can submit relief requests including details such as relief type, location, household information, and severity level.

Main System Pages:

• Registration Page

• Login Page

• User Dashboard

• Relief Request Submission Form

• Admin Dashboard

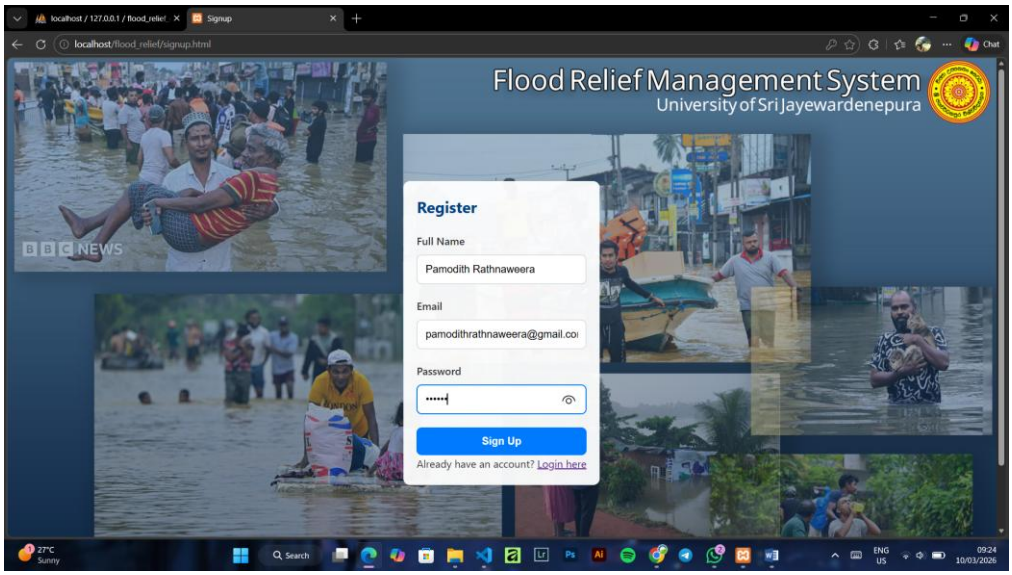
• Summary Report Page

Screenshots of the above system components should be inserted in this section to demonstrate how the application works.

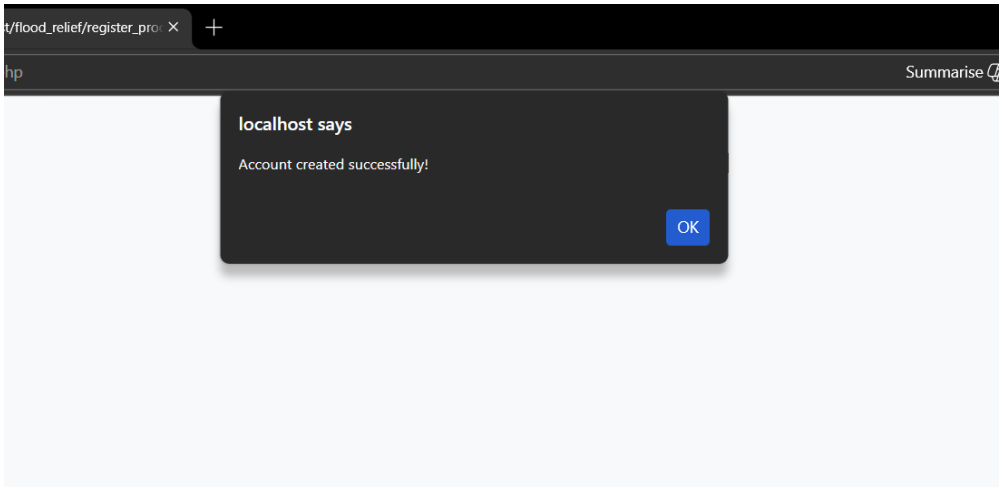
Screenshots

- **Outputs of the user interface**

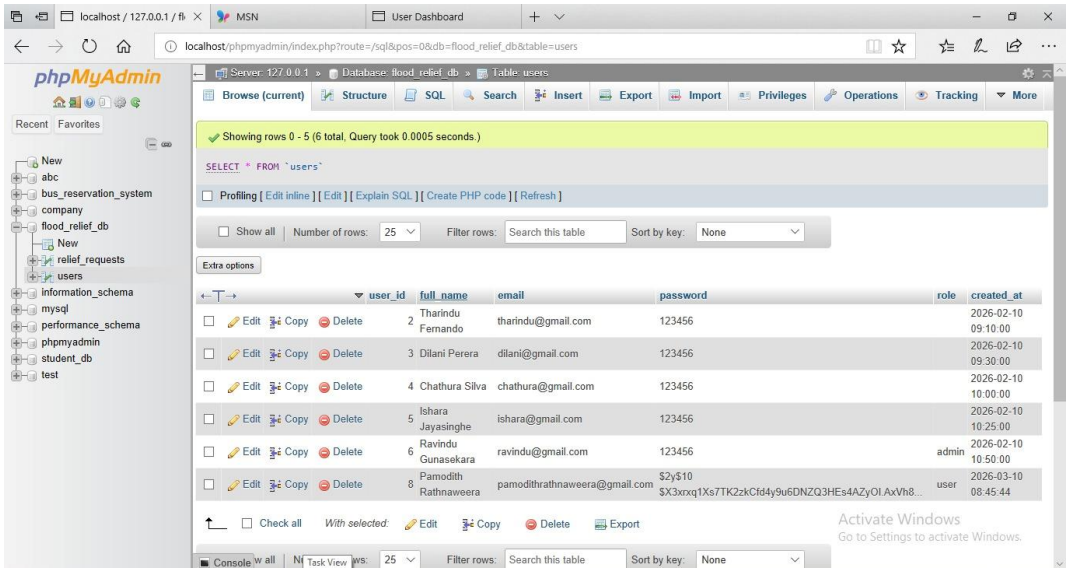
Registration Page:



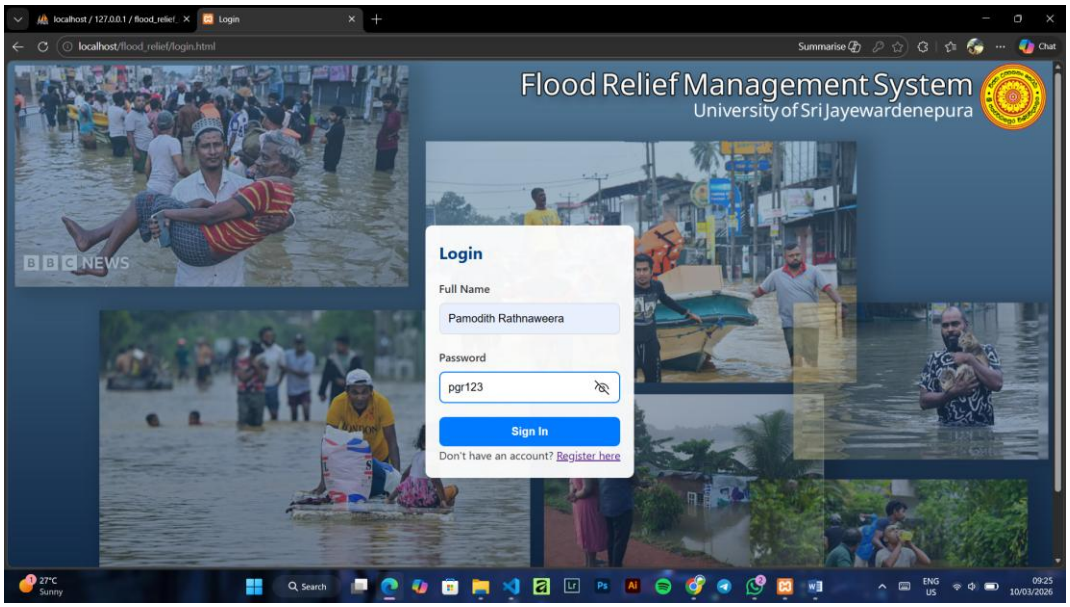
Account Created Successful Alert:



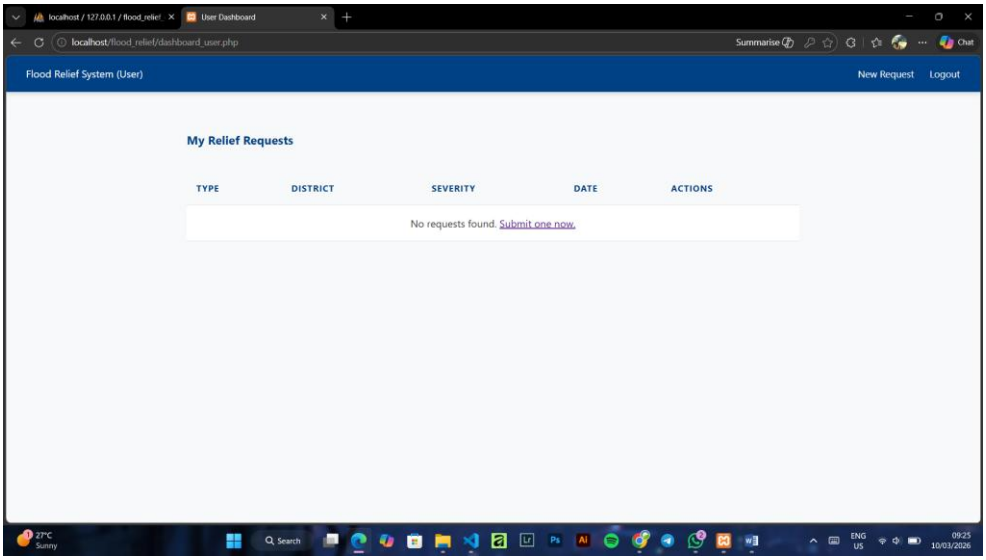
Data base after user registration:



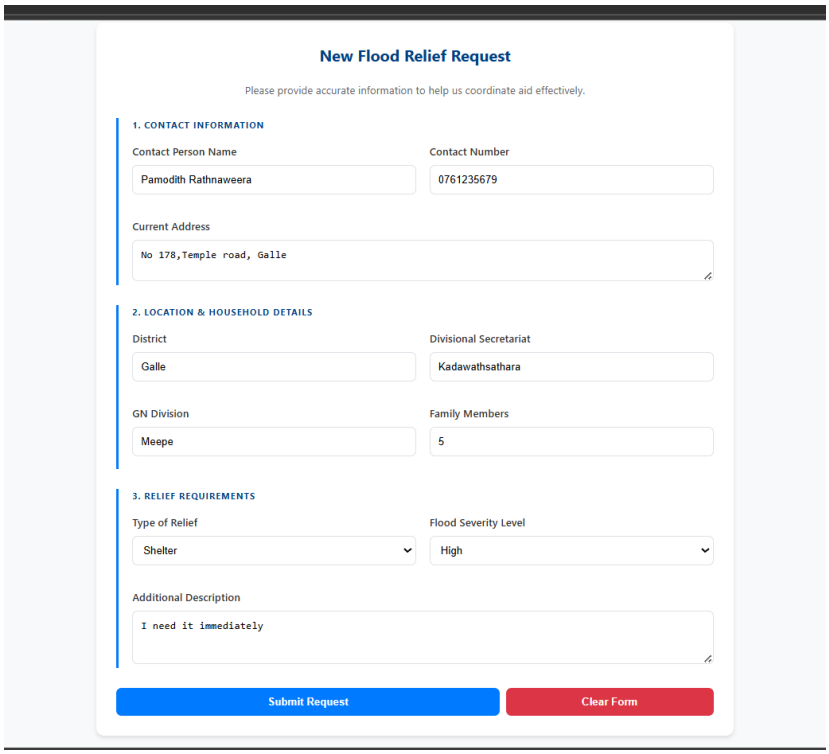
User Login Page:



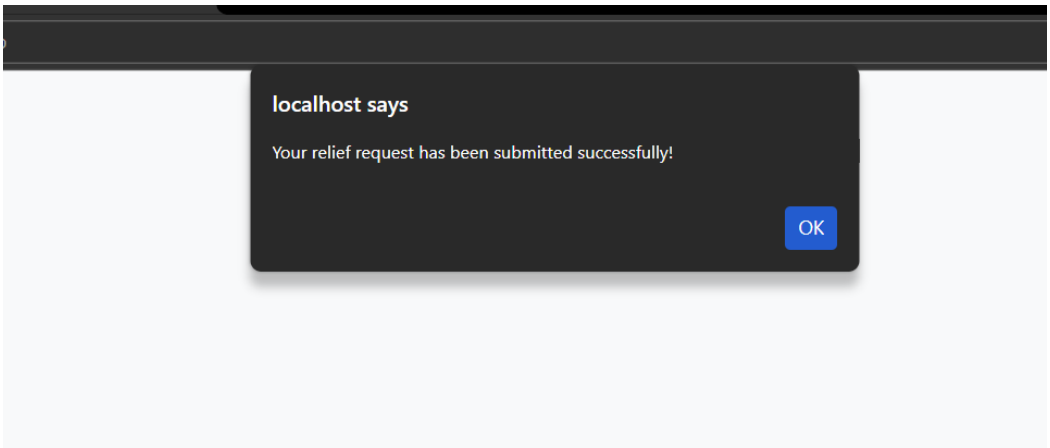
User Dashboard:



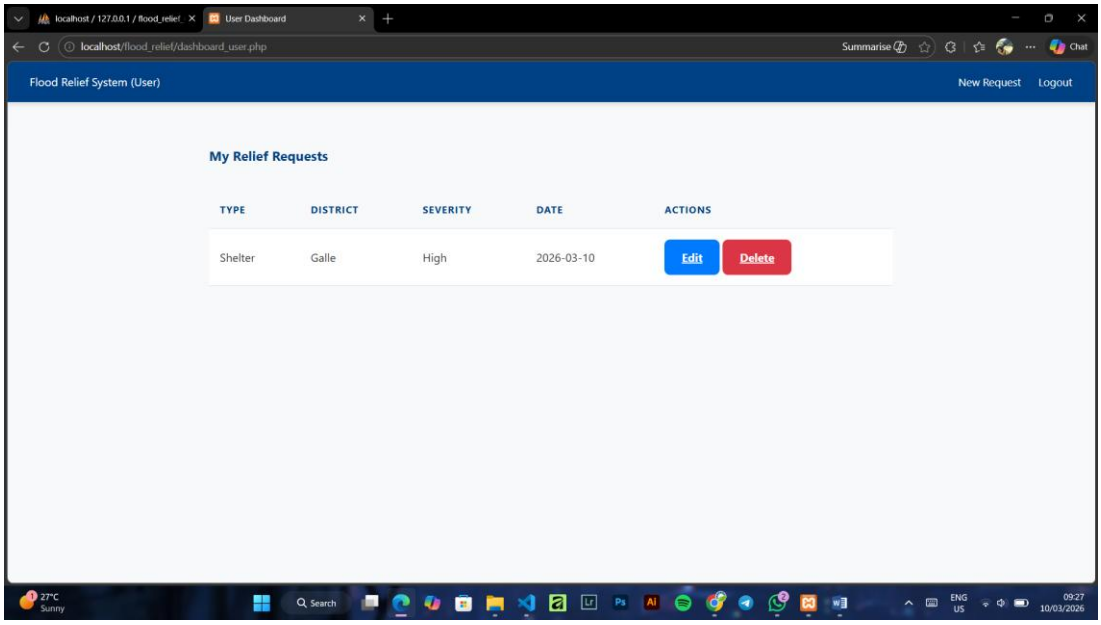
Relief Request Submission Form:



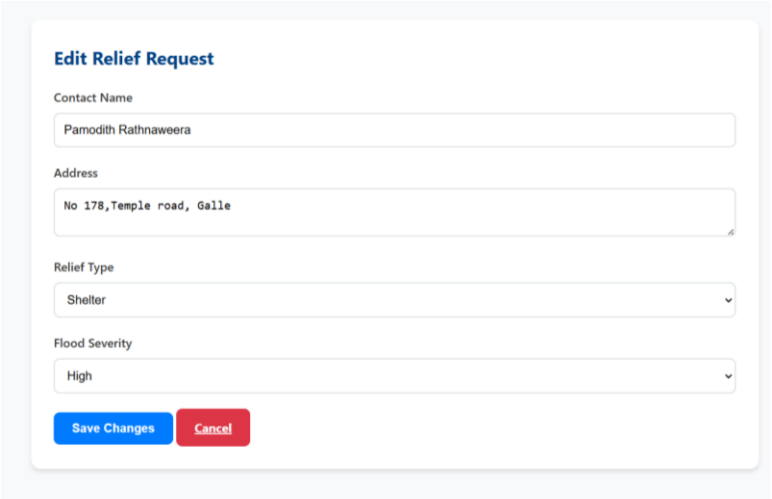
Request Form Submitted Successful Alert:



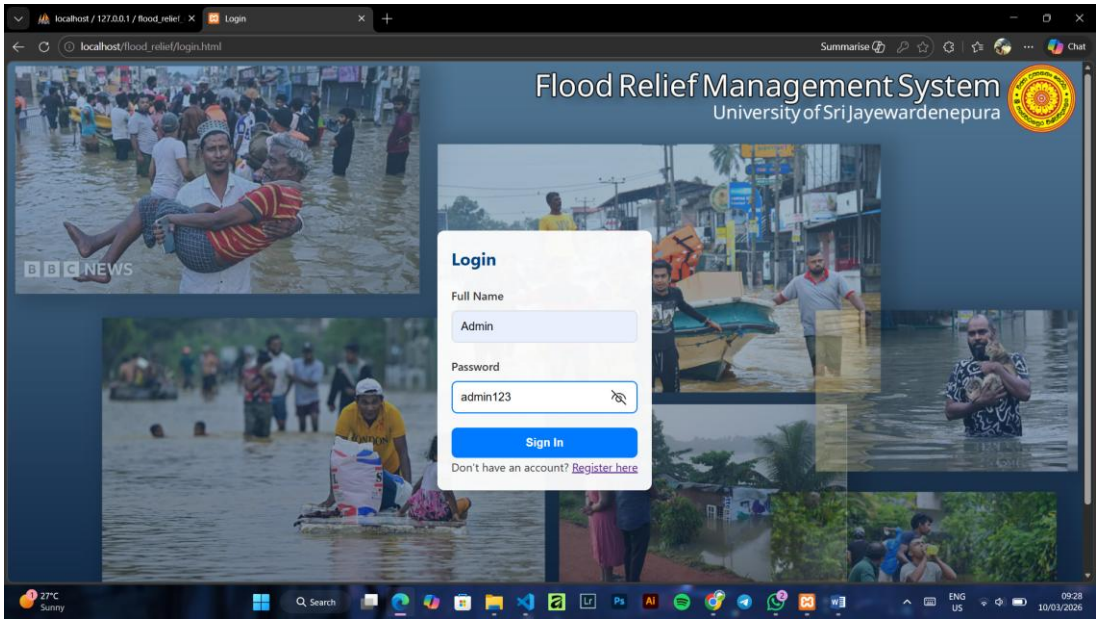
View Relief Requests:



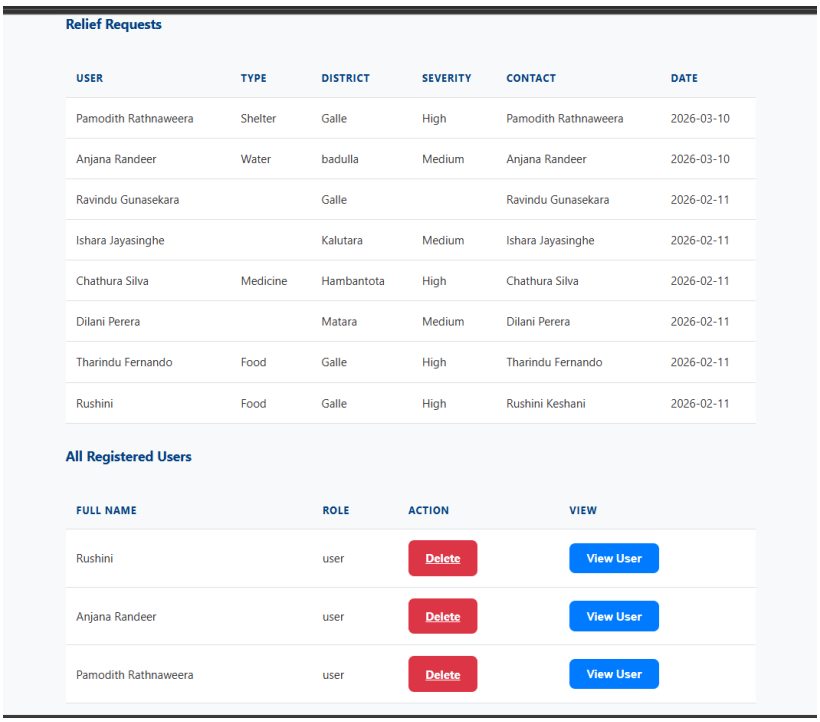
Edit Relief Request:



Admin Login:



Admin Dashboard:



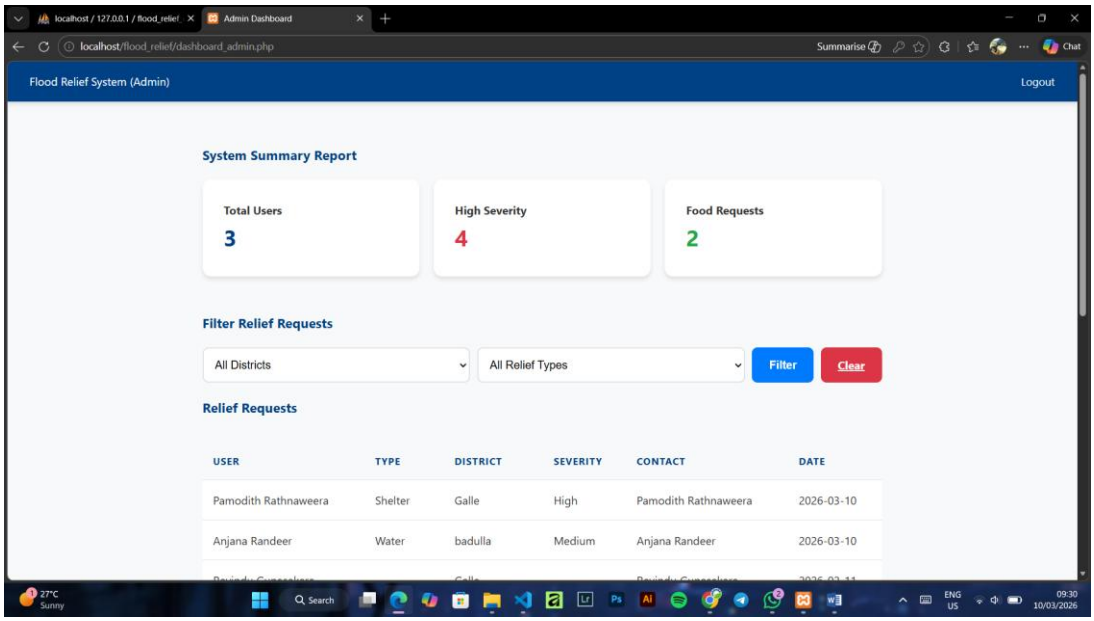
Relief Requests

USER	TYPE	DISTRICT	SEVERITY	CONTACT	DATE
Pamodith Rathnaweera	Shelter	Galle	High	Pamodith Rathnaweera	2026-03-10
Anjana Randeer	Water	badulla	Medium	Anjana Randeer	2026-03-10
Ravindu Gunasekara		Galle		Ravindu Gunasekara	2026-02-11
Ishara Jayasinghe		Kalutara	Medium	Ishara Jayasinghe	2026-02-11
Chathura Silva	Medicine	Hambantota	High	Chathura Silva	2026-02-11
Dilani Perera		Matara	Medium	Dilani Perera	2026-02-11
Tharindu Fernando	Food	Galle	High	Tharindu Fernando	2026-02-11
Rushini	Food	Galle	High	Rushini Keshani	2026-02-11

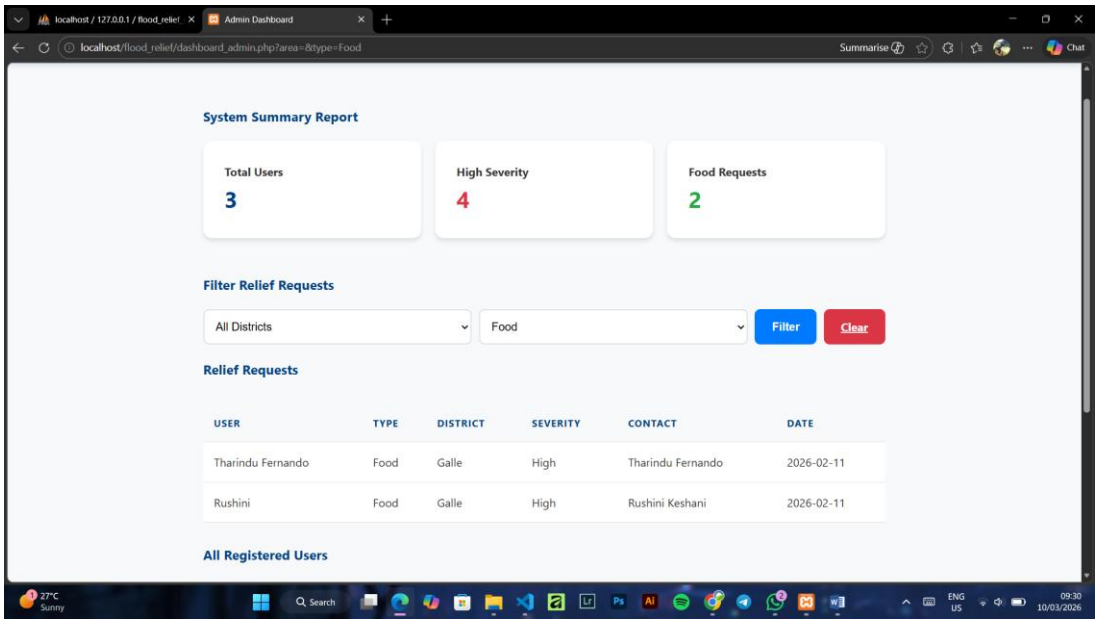
All Registered Users

FULL NAME	ROLE	ACTION	VIEW
Rushini	user	Delete	View User
Anjana Randeer	user	Delete	View User
Pamodith Rathnaweera	user	Delete	View User

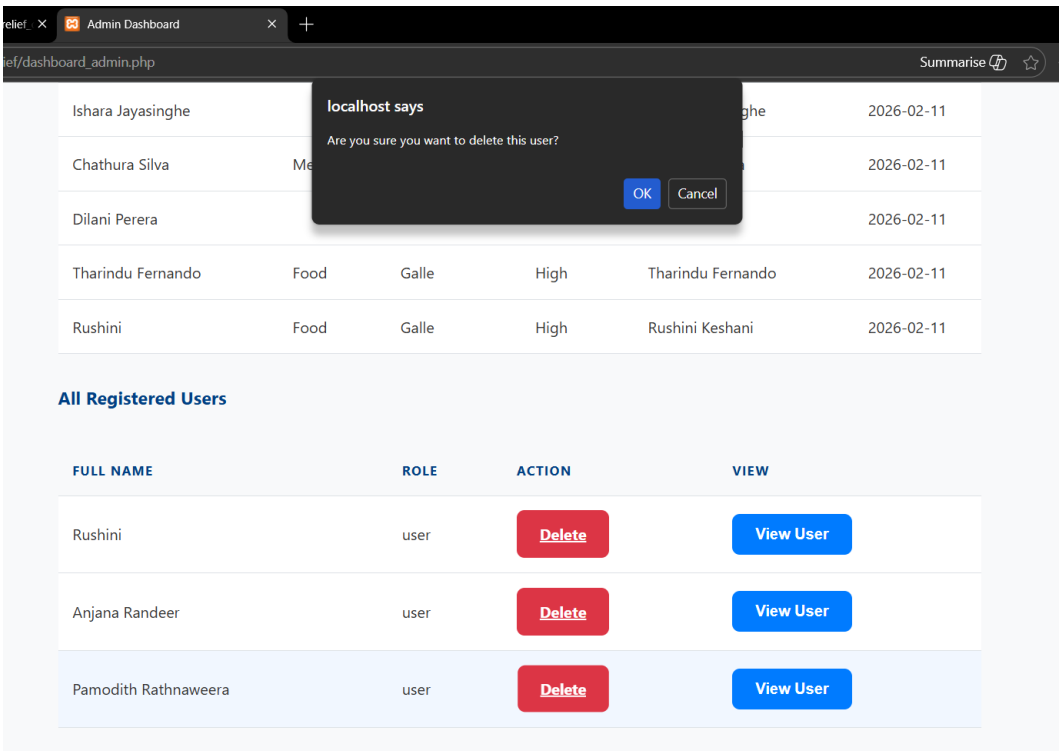
System Summary Report:



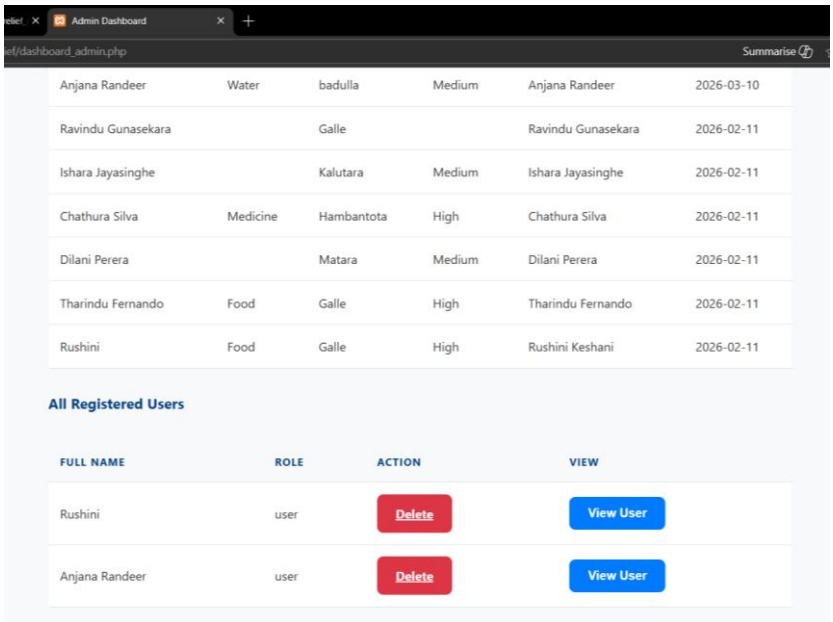
Filter Relief Requests:



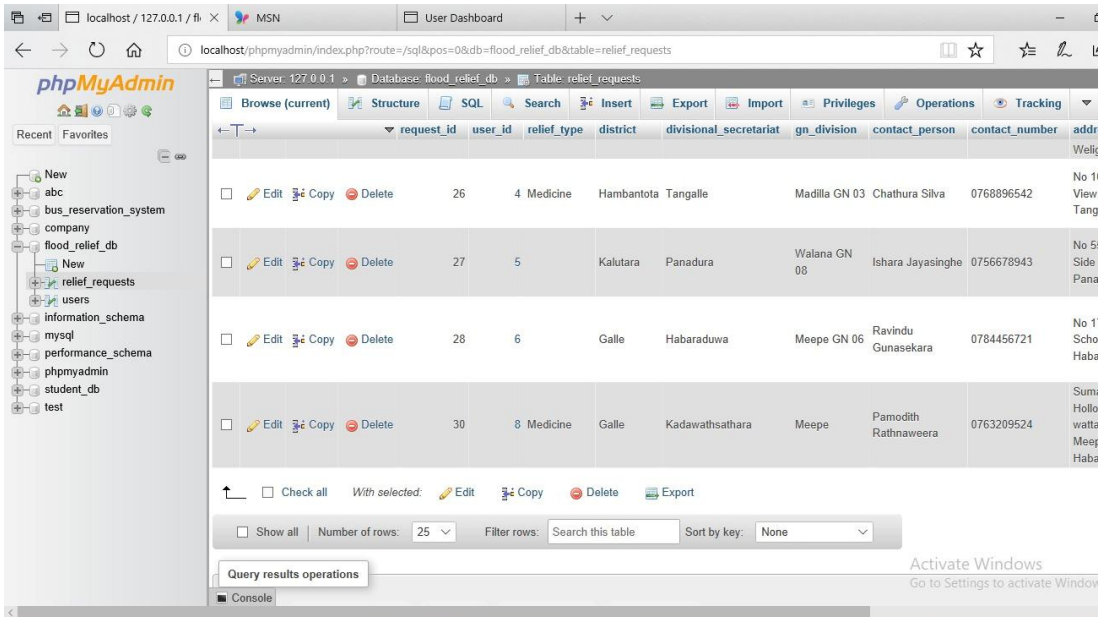
User Delete Confirmation:



After Delete User:

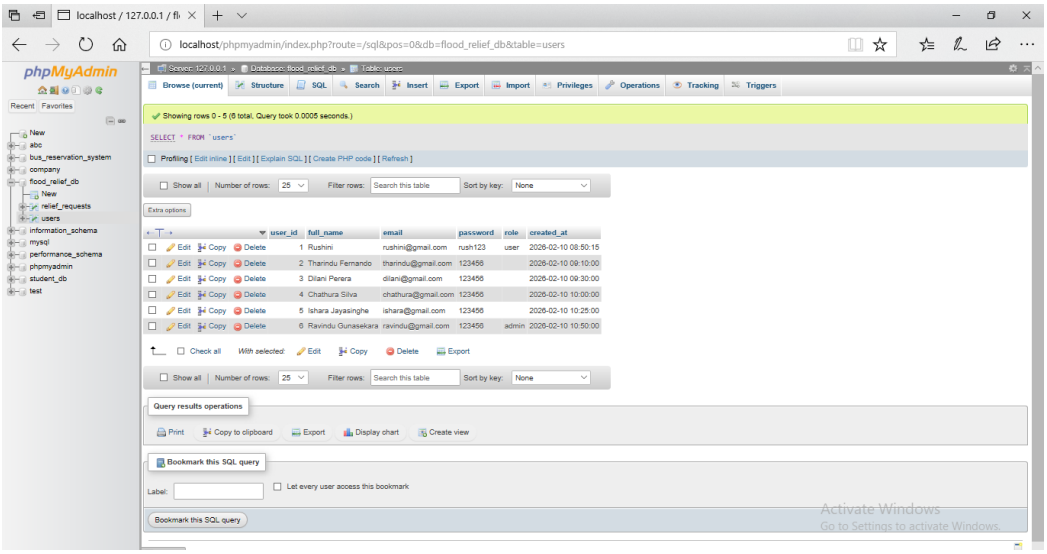


Database after delete Relief Request:

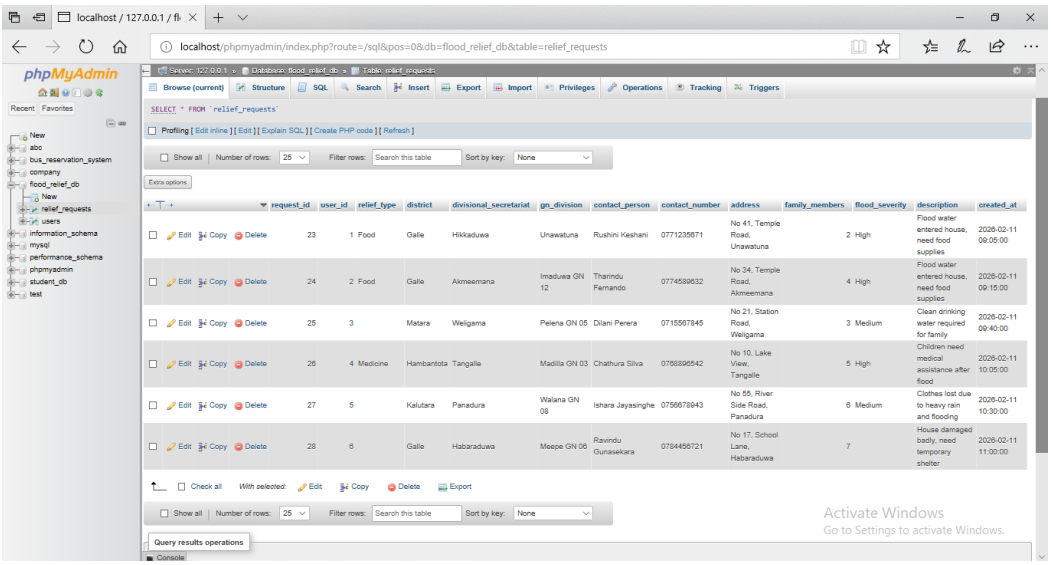


- **SQL Queries**

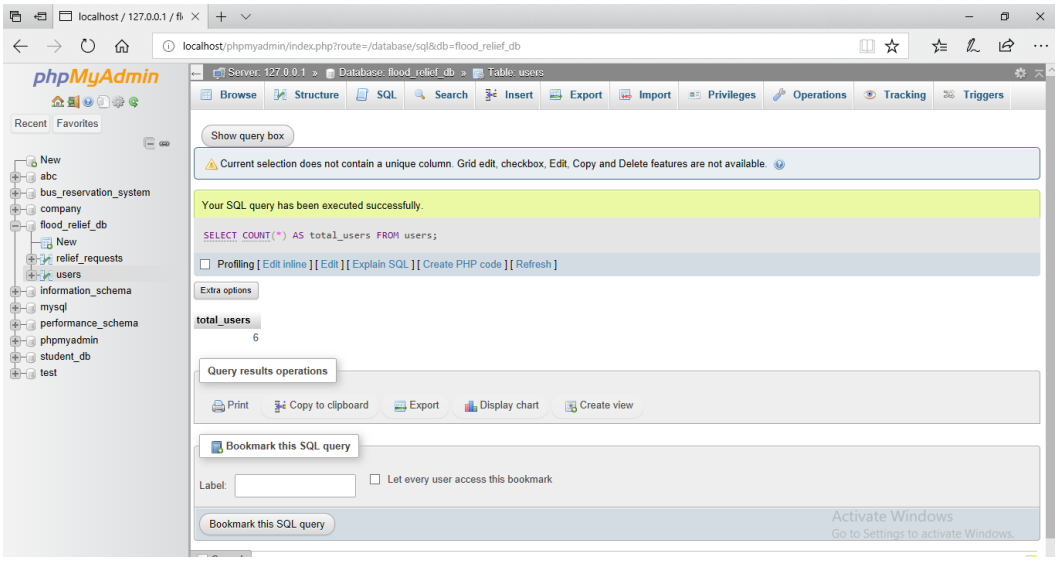
User Table:



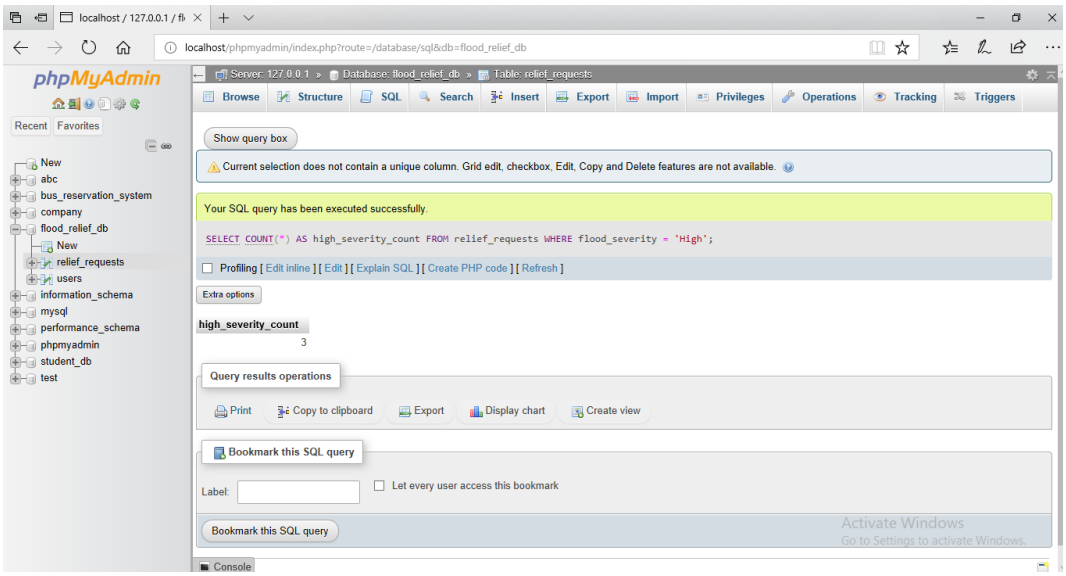
Relief Request Table:



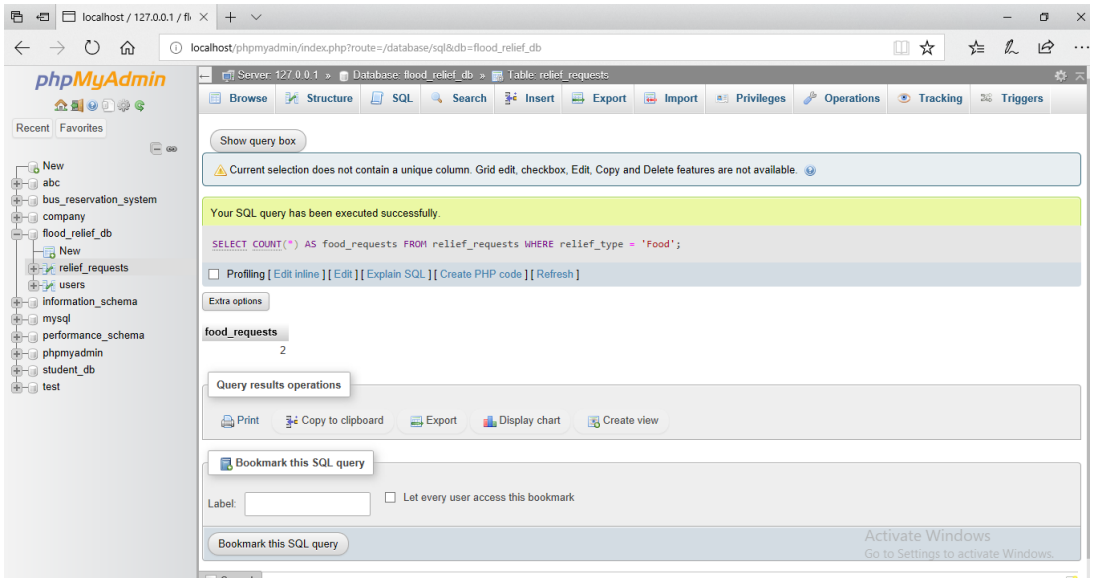
Total Registered Users:



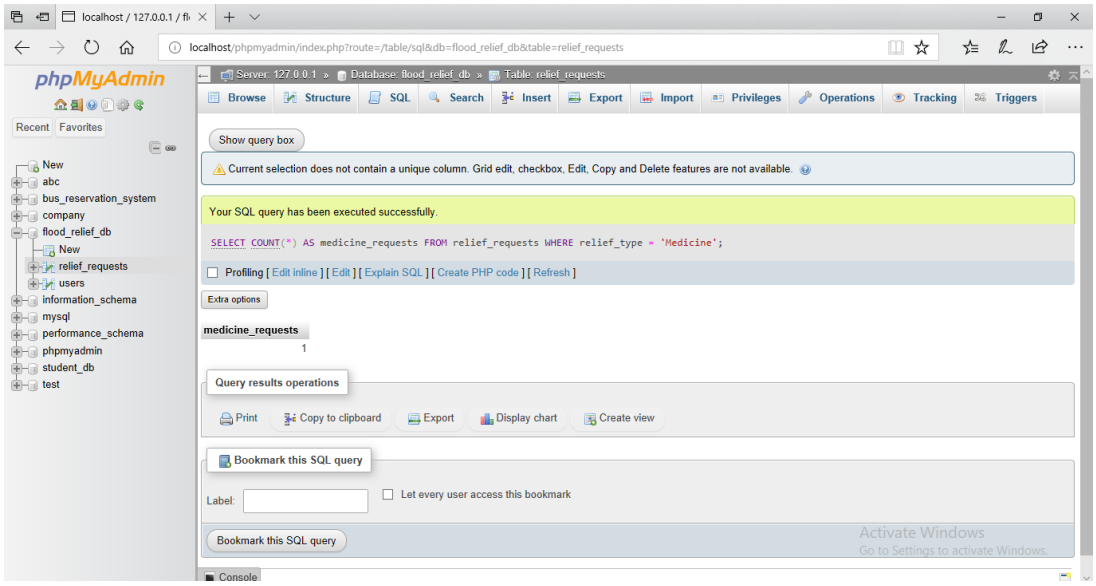
Total Food Requests:



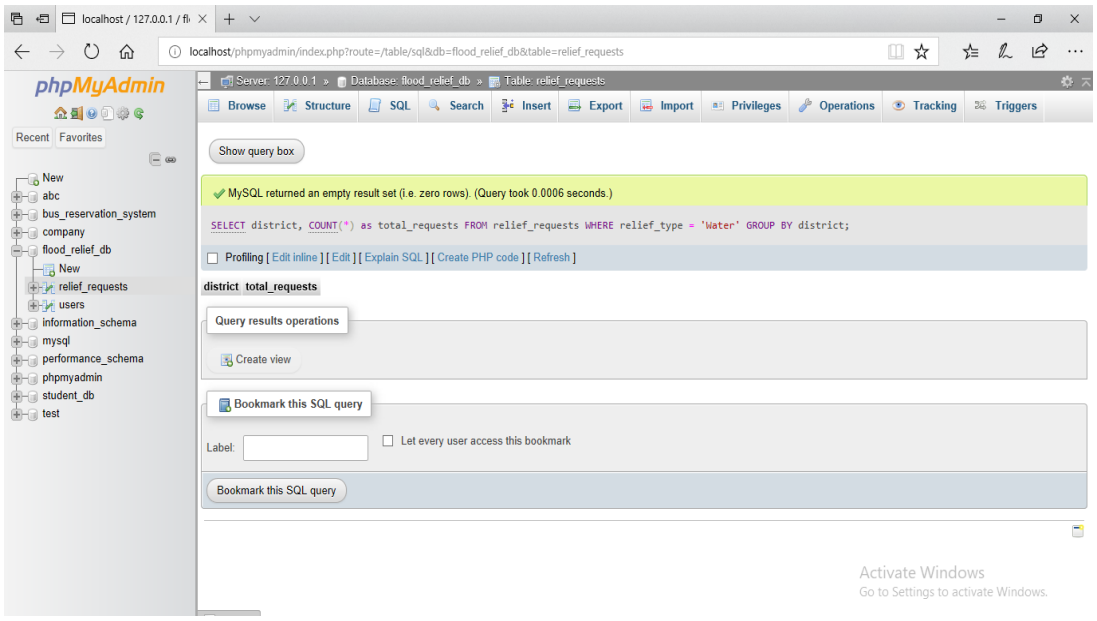
Total Medicine Requests:



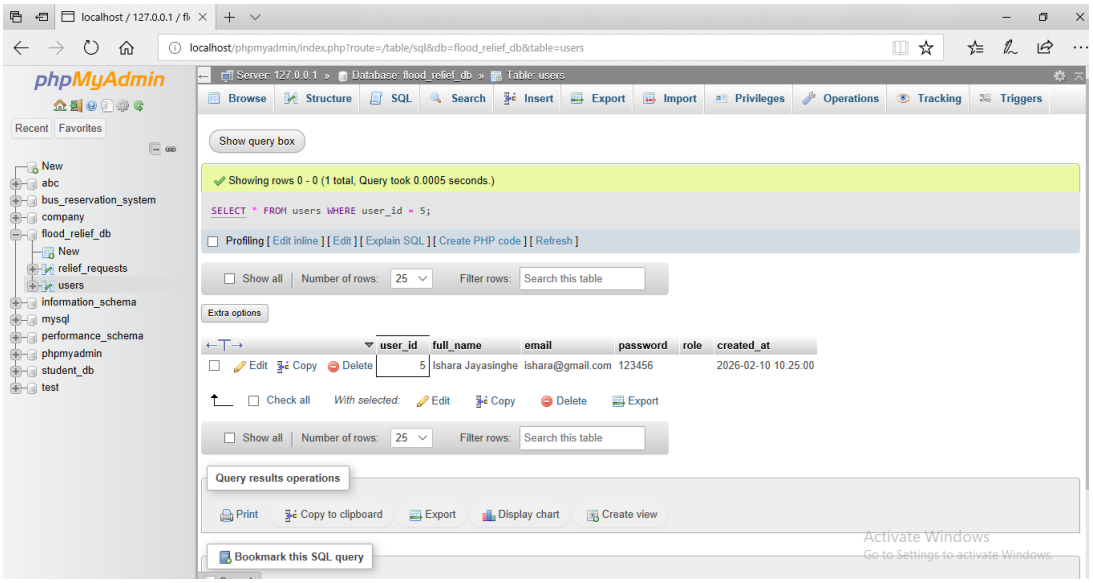
High Severity Households:



Count Specific Relief Per District:



View Detailed Information for a single user:



The Flood Relief Management System is a web-based platform designed to streamline the collection and management of relief requests for communities affected by natural disasters in Sri Lanka. Developed as a full-stack application, it connects affected individuals with administrators to ensure efficient aid distribution. Overall, the project successfully demonstrates the use of full-stack development concepts including database connectivity, server-side processing, and user interface design.

Git Hub repository-link-https://github.com/Anjana-Randeer/Flood-Relief-Management-System-Group-18-.git

4. Group Members' Contributions

Each group member contributed to different parts of the project development.

Member 1: AS20240389 – Dihini Karannagoda

- Backend development

Member 2: AS20240626 – Anjana Randeer

- Frontend interface design – HTML & CSS

Member 3: AS20240593 – Rushini Keshani

- Database design
- Documentation and report preparation

Member 4:AS20240536 – Pamodith Rathnaweera

- Frontend – JavaScript
- Documentation and report preparation

Signatures:

1. AS20240389	
2. AS20240626	
3. AS20240593	
4. AS20240536	

Thank you!